

Department of Computer Applications
23MX21 – Software Project Estimation
Learning Outcomes based questions

1. Apply COCOMO technique to compute all the estimates required for an embedded project having 4 modules of size 12.34 KLOC, 12,345 LOC, 33.125 KLOC and 23,678 LOC respectively. Consider the following cost drivers: High Skilled Programmer and Analyst, High Database Size and Very High Use of Software Tools.
2. Suppose you are developing a software product of organic type. You have estimated the size of the product to be about 100000 lines of code. Compute the nominal effort and the development time.
3. A semidetached project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the Effort, development time, average staff size.
4. A new project with estimated size of 40,000 LOC embedded system has to be developed. Project Manager of this project has a choice of hiring from two groups of developers A & B.
 - a. Group A: Very high application experience, very little experience in the programming Language and very high skilled Analysts.
 - b. Group B: Very low application experience, very high experience in the programming Language and very low skilled Analysts.

Analyze and suggest to Project Manager with evidence, the impact of hiring the developers either from Group A or B. $a=2.8$, $b=1.20$, $c=2.4$, $d=0.32$.

5. Is it true that a software product can always be developed faster by having a larger development team? Assume that all developers are equally skilled & experienced.
6. Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded.
7. Suppose you are project manager of a large project. The top management informs that you would have to do with a fixed team size throughout the duration of project. What will be the impact of this decision on your project?

8. Consider an Organic project and it is decomposed into 7 distinct functions. Apply decomposition techniques to compute total number of lines of code, cost, time and effort required to complete the project. Use the following empirical equation to propose the LOC values: $C_1 + C_2F$, where C_1 and C_2 are constants and F is the Function number. Use the following constant table to propose three LOC values. Assume that 2% of m will be paid as \$/Loc. and 7% of b will be used to propose number of lines per month.

Constants/LOCs	a	m	b
C_1	351	373	450
C_2	624	692	737

9. If a software product of size S takes m months to develop, then according to the COCOMO estimation model, how long (in months) will it take to develop a product of size $2 \times S$?
- a) Greater than $2 \times m$ months
 - b) Greater than $3 \times m$ months
 - c) Less than $2 \times m$ months
 - d) Greater than $4 \times m$ months